

Testing NS8 modules on real nodes with GitHub Actions

A free CI pattern born out of BigBlueButton's own update cadence

Where this started

The problem

- BigBlueButton module: 17 container images plus UI dependencies
- Renovate opens dependency PRs continuously
- No way to know an update still works before someone installs it
- The existing test workflow needs Nethesis-owned infrastructure (DigitalOcean, NS8-CI project, ci.nethserver.net), never run outside Nethesis

The idea

- Boot a real NS8 node under KVM, directly on GitHub's own runners
- No Nethesis infrastructure, no secrets to manage
- Works from a fork or a pull request
- Free and unmetered on public repositories

How it works: three nested environments

GitHub runner: ubuntu-24.04, ephemeral

bridge ns8br0: MASQUERADE → ghcr.io, docker.io

QEMU / KVM guest: real NS8 node

192.168.77.10

guest distro: debian12, debian13 or rocky9

ns8-core: traefik :80 :443 → module pod

Test container

netns = host

test-module.sh
→ robot framework

connects over SSH



The runner is destroyed at the end of the job. The bridge and the built image disappear with it.

Three phases, one job

1

Resolve the image

skopeo resolves the digest before anything boots. The summary names the exact artifact under test.

2

Bring the node up

Cloud-init seeds the guest; over SSH: install.sh from ns8-core, then create-cluster.

3

Run the suite

test-module.sh starts the Robot container: add-module → the module's own assertions → remove-module.

What actually gets asserted

From tests/bigbluebutton.robot, real Robot Framework cases running against a live node:

- ✓ Check if the port ranges were allocated
- ✓ Check if the mediasoup range is the expected size
- ✓ Check if the secrets file is not world readable
- ✓ Check if configure-module refuses a missing public address
- ✓ Check if FreeSWITCH answers on the event socket
- ✓ Check if nginx answers behind Traefik
- ✓ Check if bigbluebutton is removed correctly

Proof: a real Renovate run

17m 11s

workflow duration
(2 distros in parallel)

2

distros tested
(Rocky 9 + Debian 13)

Success

job status

2

test-output artifacts
published

Triggered by a Renovate PR bumping a vue/axios lockfile. Caught and validated before merge, not after install.

Each run tags and digests the exact image tested, and packages the Robot log, the guest journal and the QEMU console for debugging a failure.

github.com/stephdl/ns8-bigbluebutton, run #5, test-module-qemu.yml

Cost, and where it stops

Cost

- Free and unmetered: standard GitHub-hosted runners, public repository
- One added file per module (~46 lines), reusing build-images.sh / test-module.sh / tests/
- Guest distro is a simple choice: debian12, debian13 or rocky9
- Cloud image cached on its published checksum: about 290s on a cold Rocky pull, about 5s on a cache hit
- A full run takes about ten minutes

Limits

- Concurrent jobs capped by plan: 20 on Free, 40 on Pro
- About 12 GiB is the practical RAM ceiling per guest
- Larger, non-standard runners are always billed, even on public repos
- Never tried on a self-hosted runner: the bridge, tap and NAT rules are not cleaned up between jobs

**Started as a workaround
for one module's update cadence.**

*Ended up a reusable, infrastructure-free CI pattern
any NS8 module can adopt with one file.*